Final exam

① Determine if the vectors $\begin{bmatrix} 1 \\ -2 \\ 3 \end{bmatrix}, \begin{bmatrix} 3 \\ 2 \\ 1 \end{bmatrix}, \begin{bmatrix} 5 \\ 6 \\ -1 \end{bmatrix}$ are linearly independent or linearly dependent. Please explain your answer.

② Let $A = \begin{bmatrix} 1 & 2 & -3 \\ 1 & 0 & 2 \\ -3 & 4 & 6 \end{bmatrix}$. For which vectors b does the equation $Ax = b$ have a solution?

③ Let A, B be square matrices of the same size such that A is NOT invertible (B may or may not be invertible). Explain why the matrix AB is not invertible.

④ If A is an 8×5 matrix, find the smallest possible value of $\dim \operatorname{Nul} A^T$.

⑤ Find eigenvalues and eigenvectors of the matrix

$$\begin{bmatrix} -14 & 12 \\ -20 & 17 \end{bmatrix}.$$

(continued on the next page)

(6) Give an example of a 2×2 Matrix A and two vectors u and v such that u and v are orthogonal to each other, but Au and Av are not orthogonal to each other.

(7) Apply Gram-Schmidt process to the vectors

$$\begin{bmatrix} 1 \\ -3 \\ 5 \end{bmatrix}, \begin{bmatrix} 2 \\ 2 \\ 1 \end{bmatrix}$$

(8) (Extra Credit) Let A be an $n \times n$ matrix. If, for some vector b, the equation $Ax = b$ has more than one solution, explain why A is not an invertible matrix.